



PTT PHENOL COMPANY
LIMITED

PROCESS SPECIFICATION

DECOMPOSER COOLER (E-2307)

Docu. No. : 14780-8120-PS-E2307
Project No. : 120117
Rev. No. : Z1
Date : MAY 11, 2016
Page : 1 / 8

TITLE :

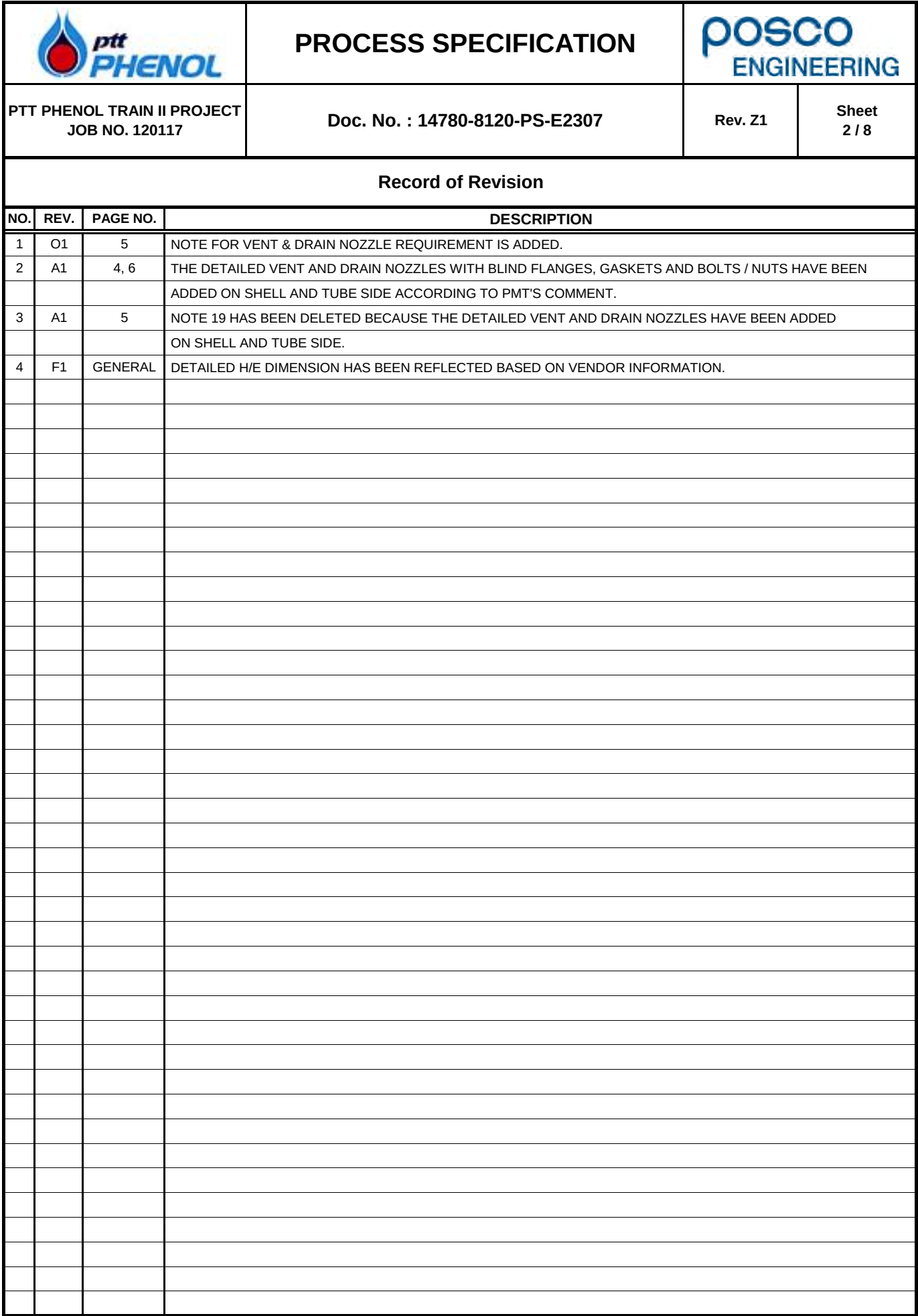
DECOMPOSER COOLER

(E-2307)

DISCIPLINE : **PROCESS**
DOCUMENT NO. : **14780-8120-PS-E2307**
PROJECT NAME : **PTT Phenol Train II Project**
PROJECT NO. : **120117**
OWNER : **PTT Phenol Company Limited (PPCL)**

Z1	MAY 11, 2016	AS BUILT	S.W.Park	J.S.Chung	-	S.W.Lee
F1	MAY 28, 2014	FINAL ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
A1	OCT 8, 2013	APPROVED FOR DESIGN	S.W.Park	J.S.Chung	-	S.W.Lee
O1	JUN 27, 2013	COMMENT ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
REV. NO.	DATE	DESCRIPTION	PREP'N	REVIEW	QA REVIEW	APPROVAL
REV. NO.	PREPARATION DEPT.	PREP'N	REVIEW	QA REVIEW	APPROVAL	
Z1	PROCESS	S.W.Park	J.S.Chung	-	S.W.Lee	
		MAY 11, 2016	MAY 11, 2016	-	MAY 11, 2016	

This Document is the property of POSCO Engineering Co., Ltd. Therefore, it shall not be released to Any third party without permission of an authorized personnel of the POSCO Engineering Co., Ltd.





SHELL AND TUBE EXCHANGER THERMAL DATA SHEET

System of Units **Metric**

Doc. No. : 14780-8120-PS-E2307

Rev. No. : Z1

Item No. : E-2307A/B

Page : 3 / 8

Case		Design	TEMA Type	AEU	Orientation	Horizontal	Rev
Number of Units		1	Shells per Unit	2	Shells in Series	1	
Surface per Unit	m ²	1603.4	Surface per Shell	m ² 801.7	Shells in Parallel	2	
PERFORMANCE OF ONE UNIT							
		In	SHELLSIDE	Out	In	TUBESIDE	Out
Fluid Name		Reliable Cooling Water			Phenol, Acetone, Cumene, H ₂ SO ₄ , H ₂ O		
Total Flowrate (Note 2)		kg/h	3349751 × 1.1		kg/h		1826205 × 1.1
Vapour		kg/h	0	0	kg/h		0
Liquid		kg/h	3349751 × 1.1	3349751 × 1.1	kg/h		1826205 × 1.1
Fluid Condensed		kg/h	0		kg/h		0
Fluid Evaporated		kg/h	0		kg/h		0
LIQUID PROPERTIES							
		AS WATER		SEE PAGE 7			
Density	kg/m ³						
Viscosity	cP						
Specific Heat	kcal/kg.°C						
Thermal Conductivity	kcal/h.m.°C						
Surface Tension	dyne/cm						
Pour Point Temperature	°C						
VAPOUR PROPERTIES							
Density	kg/m ³						
Viscosity	cP						
Specific Heat	kcal/kg.°C						
Thermal Conductivity	kcal/h.m.°C						
Hydrate Temperature	°C						
THERMAL AND HYDRAULIC DATA							
Operating Temperature	°C	37.4	43	78.7	58.4		
Operating Pressure	kg/cm ² g	4.0		3.69			
Number of Passes per Shell		1		2			
Average Velocity	m/s	1.16	(Note 14)	1.41			
Pressure Drop (Allow / Calc)	kg/cm ²	0.703 (Note 17)	0.874	0.984	0.319		
Fouling Factor	h.m ² .°C/kcal	0.0004		0.0006			
Duty (Note 2)	kcal/h × 10 ⁶	18.59 × 1.1		MTD	°C	27.1	
U-value Service/Clean	kcal/h.m ² .°C	477	1031	Over Surface	%	1.5	
DESIGN CONDITIONS							
		SHELLSIDE		TUBESIDE			
Design pressure max/min	kg/cm ² g	13.0		16.5			
Full Vacuum		Not Required		Required		at 79 °C	
Test Pressure	kg/cm ² g	To Code		To Code			
Design Temperature max/min	°C	120	15	250	15		
Corrosion Allowance	mm	3.2		1.5			
Insulation Details		None		Personnel Protection			
Hydrogen Service		No		No			
Sour Service		No		No			
Lethal Service		No		No			
Cyclic Service		No		No			
Cleaning Method		Mechanical		Mechanical			
Stress Relief for Process Reasons		No		No			
Heat Release Curve		No		Yes			



SHELL AND TUBE EXCHANGER THERMAL DATA SHEET

System of Units Metric

Doc. No. : 14780-8120-PS-E2307

Rev. No. : Z1

Item No. : E-2307A/B

Page : 4 / 8

CONSTRUCTION OF ONE SHELL

Rev

SHELL	Shell I.D.	mm	1600	Channel I.D. (If different)	mm	
TUBES	Number of Tubes		2238 (Note 3)	Tube Length	mm	6096
	Tube O.D.	mm	19.05	Tube Thickness	mm	1.651
	Tube Gauge	BWG	16	Tube Wall Basis		Average
	Tube Pitch	mm	25.4	Tube Layout Angle	deg.	90
	Tube-Tubesheet Joint		Strength Welded (Note 4)	Tube-Baffle Diam. Clearance	mm	0.8
BAFFLES	Number of Baffles		6	Baffle Type		NTIW
	Baffle Cut	%	16	Baffle Spacing	mm	650
	Baffle Cut Orientation		Horizontal	Double Seg. Baffle Overlap	mm	0
	Baffle Thick	mm	12.7	Baffle-Shell Diam. Clearance	mm	7.9
BUNDLE	Impingement Protection		Rods (Note 6)	Tube Supports (Note 5)		U-bend Tan Line
	Pairs of Sealing Strips		6	Sealing Strip Details (Note 8)		Maximum Length
	Passlane Sealing		No	Passlane Sealing Details		None
	U-Bend Diam.	mm	57.15	U-Bend Orientation		Vertical
	U-Bend Area Included		No	U-Bend Support Details		(Note 7)
	Longitudinal baffle		N.A.	Longitudinal Baffle Seal		N.A.

CONNECTIONS

SHELLSIDE

Service	No.	N.B.		Rating	Facing
Inlet S1	1	24	Inch	150	RFWN
Outlet S2	1	24	Inch	150	RFWN
Vent SV	1	2	Inch	150	RFLWN
Drain SD	1	2	Inch	150	RFLWN

TUBESIDE

Service	No.	N.B.		Rating	Facing
Inlet T1	1	16	Inch	300	RFWN
Outlet T2	1	16	Inch	300	RFWN
Vent TV	1	3/4	Inch	300	RFLWN
Drain TD	1	3/4	Inch	300	RFLWN

MATERIALS

Tubes	304L Stainless Steel (Seamless)
Shell / Shell Cover	Killed Carbon Steel (UOP Std 3-11)
Floating Head Cover	N.A.
Channel / Channel Cover	304L Stainless Steel (UOP Std 3-15)
Tubesheet, Stationary	304L Stainless Steel
Tubesheet, Floating	N.A.
Baffles / Tube Supports	304 Stainless Steel
Nozzles / Nozzle Flange	Shellside: Killed Carbon Steel
Nozzles / Nozzle Flange	Tubeside: 304L Stainless Steel
Bolt and nuts	Per Project Specifications
Gaskets, Shellside	Per Project Specifications
Gaskets, Tubeside	Per Project Specifications
Gaskets, Nozzles	Per Project Specifications

ESTIMATED WEIGHTS

Weights per Shell	Exchanger Weight, Dry	26.8	Exchanger Weight, Wet	43.8
(Tonnes)	Bundle Weight, Dry	13.0		

DESIGN REQUIREMENTS

PED Classification	Shellside- N.A.	Tubeside- N.A.
Project Specifications	UOP 963766 - 401, UOP Std 4-11-7, UOP Std 3-11, UOP Std 3-15, UOP Std 9-11-7, 14780-21A1	
Design Code	ASME VIII Div 1	
Equipment Code	TEMA Class R 9th edition, API 660 8th edition 2007	

SHELL AND TUBE EXCHANGER THERMAL DATA SHEET

System of Units

Metric

Doc. No. : 14780-8120-PS-E2307

Rev. No. : Z1

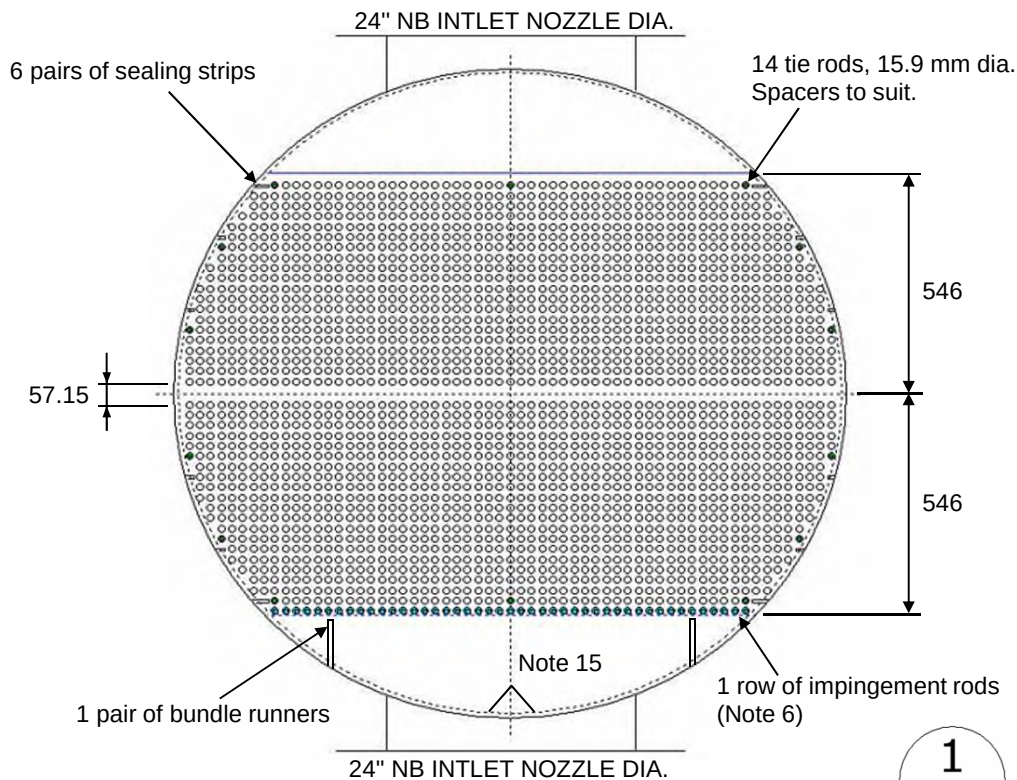
Item No. : E-2307A/B

Page : 5 / 8

Notes:

1. Item has been checked for Flow Induced Vibration. Tube damage is not predicted.
2. Flow and duty include 10% design margin.
3. For U-tube design, Number of tubes equals number of straights.
4. Strength-weld tubes to tubesheet per requirements as specified on page 8.
5. A full support plate at U-bend shall be provided.
6. Provide 1 row of impingement rods of the same diameter, same pitch, and same layout as the tubes.
7. Provide one in bend U-bend support to hold 7 outer most tube rows.
8. Sealing strips and bundle runners shall not restrict the bundle cleaning lanes.
9. Bundle runners shall not restrict the shellside entrance flow path.
10. Minimize the volume of the process material in the tube side. VENDOR to confirm.
11. The maximum allowable leakage of cooling water into the process is 100 wt ppm. VENDOR to confirm.
12. Cooling water shall be cascaded with the E-2301 Preflash and Flash Column Condensers. CONTRACTOR to confirm flow rate and inlet temperature.
13. Provide the intermediate support plates at inlet and outlet spacings. The intermediate support plate at inlet spacing shall be extended to hold the impingement rods.
14. The effective cross flow velocity is 1.97 m/s.
15. All baffles and support plates extending to the bottom of the shell shall be provided with V-notch at the lowest point.
16. A Magnesium anode shall be provided on the shellside.
17. CONTRACTOR to confirm the allowable shellside pressure drop.
18. Test ring shall be provided by VENDOR when required.
19. Deleted.

Tube Layout



O.T.L. = 1579.5 mm

All dimension in mm

Sketch not to scale

Tubepass Arrangement

Rev

SHELL AND TUBE EXCHANGER THERMAL DATA SHEET

System of Units Metric

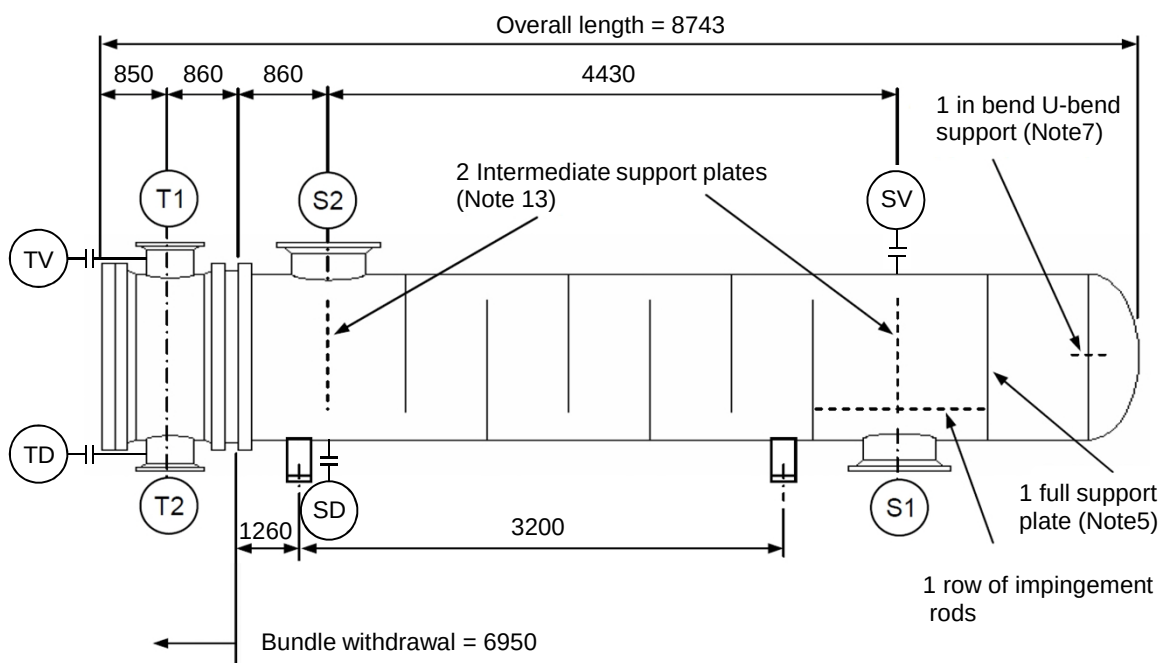
Doc. No. : 14780-8120-PS-E2307

Rev. No. : Z1

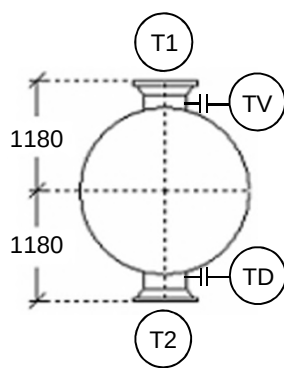
Item No. : E-2307A/B

Page : 6 / 8

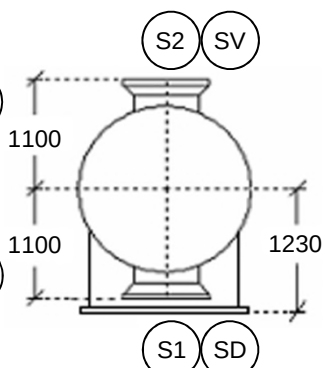
Setting Plan



Tubeside Nozzles



Shellside Nozzles



Shell ID : 1600 mm

Tube length : 6096 mm

All dimension in mm
Sketch not to scale

PROCESS NOZZLE SUMMARY

REF	SERVICE	SIZE	RATING	ELEV.	REMARK
S1	Reliable Cooling Water Inlet	24"	150#RFLWN		
S2	Reliable Cooling Water Outlet	24"	150#RFLWN		
T1	Phenol, Acetone, Cumene, H ₂ SO ₄ , H ₂ O Inlet	16"	300#RFLWN		
T2	Phenol, Acetone, Cumene, H ₂ SO ₄ , H ₂ O Outlet	16"	300#RFLWN		
SV	Shellside Vent	2"	150#RFLWN		CW/blind
SD	Shellside Drain	2"	150#RFLWN		CW/blind
TV	Tubeside Vent	3/4"	300#RFLWN		CW/blind
TD	Tubeside Drain	3/4"	300#RFLWN		CW/blind

Rev

SHELL AND TUBE EXCHANGER THERMAL DATA SHEET

System of Units

Metric

Doc. No. : 14780-8120-PS-E2307

Rev. No. : Z1

Item No. : E-2307A/B

Page : 7 / 8

HotSide Property Curve

Pressure

3.690

kg/cm²g

Temperature °C	Entire Stream Enthalpy kcal/kg	Entire Stream Mass Vapour fraction	Vapour Density kg/m ³	Vapour Viscosity cP	Vapour Therm. Cond. kcal/h.m.°C	Vapour Spec. Heat kcal/kg.°C
78.7	10.17	0.0000	-	-	-	-
76.5	9.04	0.0000	-	-	-	-
74.3	7.91	0.0000	-	-	-	-
72.1	6.78	0.0000	-	-	-	-
69.8	5.65	0.0000	-	-	-	-
67.6	4.52	0.0000	-	-	-	-
65.3	3.39	0.0000	-	-	-	-
63.0	2.26	0.0000	-	-	-	-
60.7	1.13	0.0000	-	-	-	-
58.4	0.00	0.0000	-	-	-	-

Temperature °C	Liquid Density kg/m ³	Liquid Viscosity cP	Liquid Therm. Cond. kcal/h.m.°C	Liquid Spec. Heat kcal/kg.°C	Liquid Surf. Tension dyne/cm	Critical Pressure kg/cm ² [a]	Critical Temperature °C
78.7	897.0	0.572	0.119	0.51	27.0	59.6	336
76.5	899.3	0.590	0.119	0.51	27.3	59.6	336
74.3	901.6	0.609	0.120	0.51	27.5	59.6	336
72.1	903.9	0.630	0.120	0.50	27.8	59.6	336
69.8	906.2	0.652	0.121	0.50	28.1	59.6	336
67.6	908.5	0.676	0.121	0.50	28.3	59.6	336
65.3	910.8	0.701	0.122	0.49	28.6	59.6	336
63.0	913.1	0.727	0.122	0.49	28.9	59.6	336
60.7	915.5	0.756	0.122	0.49	29.1	59.6	336
58.4	917.8	0.786	0.123	0.49	29.4	59.6	336

Rev

SHELL AND TUBE EXCHANGER THERMAL DATA SHEET

System of Units Metric

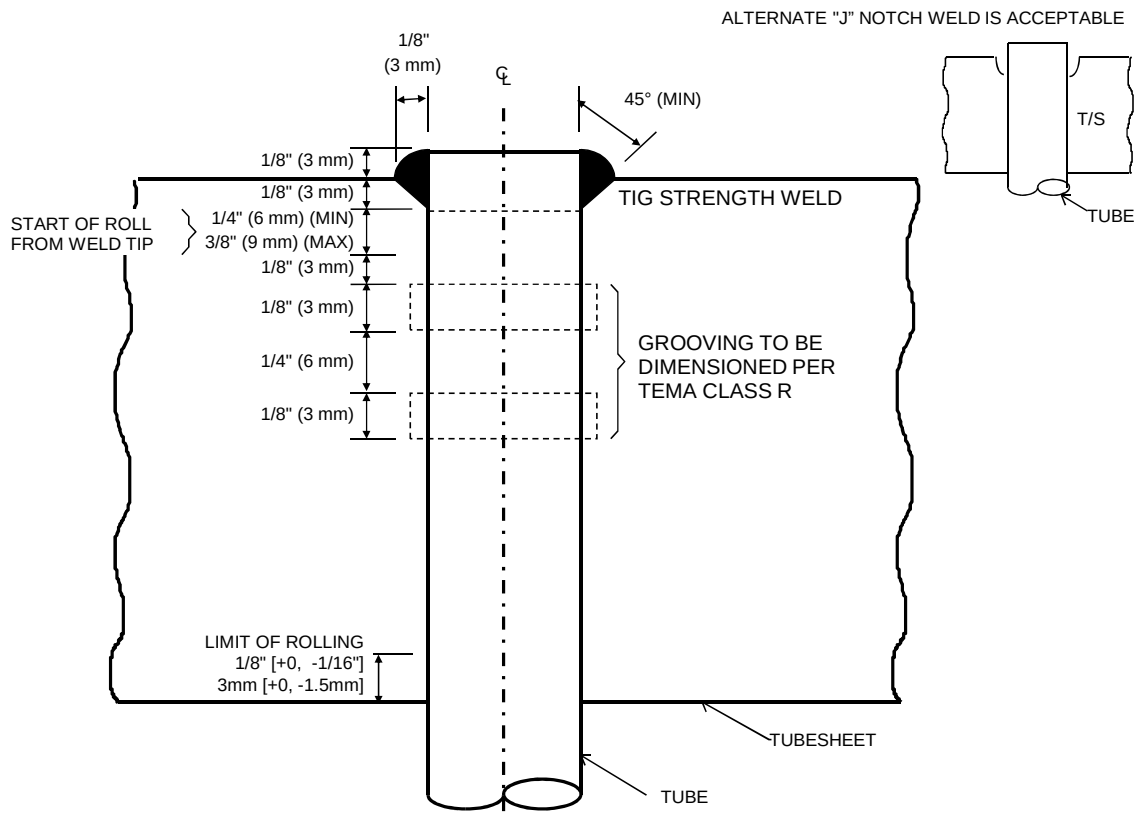
Doc. No. : 14780-8120-PS-E2307

Rev. No. : Z1

Item No. : E-2307A/B

Page : 8 / 8

Strength-Weld Type Tube to Tubesheet Joint Requirements (Refer to UOP 963766 - 401)



1. Tubing shall be seamless type.
2. Joint components shall be deburred and meticulously cleaned prior to welding.
3. When heat treatment of the tubesheet is required, final tube hole sizing shall be attained by reaming after heat treatment.
4. Preheat tubesheet to a minimum of 150°F (65°C) and maintain this temperature during welding.
5. Each tube end shall be welded individually without stopping.
6. Tube to tubesheet joints shall be welded with a minimum of two passes. The first pass shall be 100% examined by dye penetrant or magnetic particle method. Remaining passes shall be welded without interruption. The final welds shall be 100% examined by dye penetrant or magnetic particle method.
7. All tube joint welds shall be proven using a soap test [with minimum 15 psi (1.05 kg/cm²) air] prior to rolling.
8. A helium leak test shall be performed in accordance with the requirements of ASME code, Section V, Article 10. The shell side shall be pressurized with helium while the exchanger is inclined to provide good flow of helium to a single collection point on the tube side. Any leakage in excess of 1×10^{-4} standard cc/sec warrants repair.
9. Do all tube rolling after completion of welding and leak testing.
10. The tubes shall be rolled for the entire tubesheet thickness. Stop shall be used on tube roller that assures expansion is within limits given on sketch.
11. A certified test report of the final, successful helium test shall be provided and signed by an independent third party witness.
12. All applicable Weld Procedure Specifications (WPS) and Welder Qualification Records (WQR) shall be provided per clients request.

Rev